Printables by JOSEF PRUSA



OTTO-5

AK_designs

VIEW IN BROWSER

updated 16. 4. 2026 | published 16. 4. 2026

Summary

Meet OTTO-5, a friendly, expressive, modular robot inspired by Johnny-5.

3.50 hrs	1 pcs	0.20 mm	0.40 mm	PLA	100 g	Prusa CORE One / +

Hobby & Makers > RC & Robotics

Tags: robot robotics learning ottodiy otto ottorobot

OTTO-5

No disassemble! Assemble!
OTTO-5 alive! Need input! Need input!

Welcome to my HP OTTO design!

OTTO-5 is a modular, articulating, robot with expressions. It started as a concept for an articulated head, but evolved into a full robot design, allowing for more features and details. The design is inspired by Johnny-5 form the movie Short Circuit.

Just like the original Otto, this design is made with modularity in mind. OTTO-5 features many moving and articulating parts, making it fun not only to print, but also to assemble and interact with. The design is suitable for both beginners and experienced makers, with increased detail, smaller parts, and a building experience that leans toward model making and small robotics. At the same time, most parts are easy to print, keeping it accessible.

The required hardware is kept as simple as possible without making compromises, using only a few standard components such as M2 self-tapping screws, an additional servo, O-rings, and a bearing.

Features

- Turnable head (with optional extra servo)
- Head with manually adjustable expressions
- Head compatible with ultrasonic sensor
- Articulating arms and hands
- Smooth track design with bearings
- Compatible with other OTTO parts and upgrades
- Detailed Assembly Instructions

Design & Printing Details

Recommended Settings:

- **Layer height:** 0.2 mm
- **Infill:** 10–20%
- **Supports:** Only where necessary
- **Print orientation:** As provided in the files
- **Material:** PLA recommended (PETG also possible)
- **Nozzle:** 0.4 mm
- **Bed adhesion:** Not required (use a brim if needed or increase the bed temperature to 62 °C)

Most parts are designed for easy, support-free printing, though some require supports to maintain the design, ensure better quality, and keep the overall concept simple.

All parts have been printed and tested as shown in the photos. However, every printer is different, and mine is quite old and less accurate, especially when printing multiple parts at once. For better accuracy and testing, I printed one part at a time.

Estimated total print time for all parts: **31h 53m**, based on a print speed of 40 mm/s (may vary depending on your printer and settings).

Assembly instructions

In the files, you will find an assembly instructions with more information. I have tried to make the manual accessible for both beginners and more experienced users. Since I do not own an Otto robot myself and was not familiar with it before creating this design, I have not gone into detail about coding and electronics.

Required Hardware

As mentioned, the required hardware is kept as simple and widely available as possible, probably most makers will have some of the parts already. Here's a list of the needed parts:

- 1 x SG90 Micro Servo
- 4 x M2x6 self-tapping screw
- 6 x M2x8 self-tapping screw (10 x if you don't use a flanged head)
- 4 x M2x8 self-tapping screw flanged head (only recommended, not required)
- 1 x M2x16 self-tapping screw
- 4 x O-ring 60x4 (68x60x4 - NBR)
- 1 x O-ring 37x2 (41x37x2 - NBR)
- 8 x 605ZZ Bearing (5x14x5mm)

For the screws, I recommend purchasing an assortment box.

Thank You / Enjoy Printing!

I’d love to hear your feedback, and if you have any questions, feel free to reach out.

Update:

I’ve added flexible track and belt print files for anyone who can print with TPU or other flexible materials. When possible, I will add these files here as well.

Design Process & Story

When seeing the contest I was immediately drawn to its concept of making robotics and engineering accessible with a learning platform. I initially set out to design a fully articulated, motorized head, but due to size constraints, this proved impractical without major compromises, so I decided to start with a rotating head with manual articulating parts.

While searching for inspiration, I remembered Short Circuit and the robot Johnny-5. What made this character so memorable was the expression and funny personality. I wanted to incorporate that into my design, while keeping it as modular as possible like the original Otto, and possibly with other Otto design as well.

The head was designed around the ultrasonic sensors, with manually adjustable features to create different expressions, to make it really come to life. Together with the LED-lit version of the ultrasonic sensor it could be an amazing combination.

One of the biggest engineering challenges was the track system. Early designs used small interlocking links, which worked in CAD but failed when 3D printed due to scale limitations and lack of traction. To solve this, I developed a track using standard O-rings for traction, making it reliable and accessible for users without TPU printing capabilities, like myself. In the last version bearings were added to reduce friction and prevent unnecessary load on the motors.

Another important focus was printability and assembly. Parts were split and redesigned for easy access and maintenance, where possible. Snap-fit joints were designed not to only snap in place, but also to have enough resistance to hold their position during movement. Which was important so parts don’t interfere with each other.

Although the process involved many iterations, like Johnny-5, and shown on one of the photos. A lot of test parts were printed before finalizing the parts. I believe the final design balances functionality, manufacturability, and personality, and will be a great start or platform also for more experienced users. Even without the electronics, it already feels like a companion, and I hope it inspires others to explore robotics and creative engineering.

After finishing the 3d model, I made the documentation, renders, screenshots, and offcourse all the 3d printfiles. I tried to make it accessible for as well beginners and more advanced users.

This remix is based on



HP Robots | Otto Starter

by HP Robots Otto

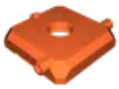
Model files

Complete Preview 1 file



otto-5_preview_noprint.stl

Top 21 files



top_base.stl



top_head_base.stl



top_head_front.stl



top_head_back.stl



top_washer_id21mm.stl



top_head_clip.stl



top_head_pully.stl



top_arm_top_r.stl



top_arm_top_l.stl



top_arm_bot_r.stl



top_arm_bot_l.stl



top_arm_vinger_r.stl



top_arm_vinger_l.stl



top_head_eyebrow_joint.stl



top_head_eyebrow_r.stl



top_head_eyebrow_l.stl



top_head_eyelid_r.stl



top_head_eyelid_l.stl



top_head_pully_cover_r.stl



top_head_pully_cover_l.stl



top_base_clean_top.stl



Middle 1 file

mid_base.stl



Bottom 16 files

bot_base.stl



bot_track_base_l.stl



bot_track_base_r.stl



bot_motorarm.stl



bot_track_wheel_large.stl



bot_track_wheel_small.stl



bot_bracket_servo.stl



bot_track_side_r.stl



bot_track_side_l.stl



bot_track_side_bracket_r.stl



bot_track_side_bracket_l.stl



bot_track_wheel_small_washer.stl



bot_track_cover_r.stl



bot_track_cover_l.stl



bot_track_base_r_open.stl



bot_track_base_l_open.stl



Accessories 3 files



acc_toolbox.stl



acc_toolbox_arm.stl



acc_toolbox_noname.stl

STEP Files 2 files



bot_track_wheel_small.stp



bot_track_wheel_large.stp



top_acc_supports.3mf



top_acc_nosupports.3mf

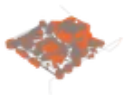


middle.3mf



bottom.3mf

Print files



bottom_example_coreone.bgcode

PLA 0.40 mm 0.20 mm 3.50 hrs 100 g Prusa CORE One / +

Example gcode for reference

Other files



manual_otto-5_a5.pdf

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